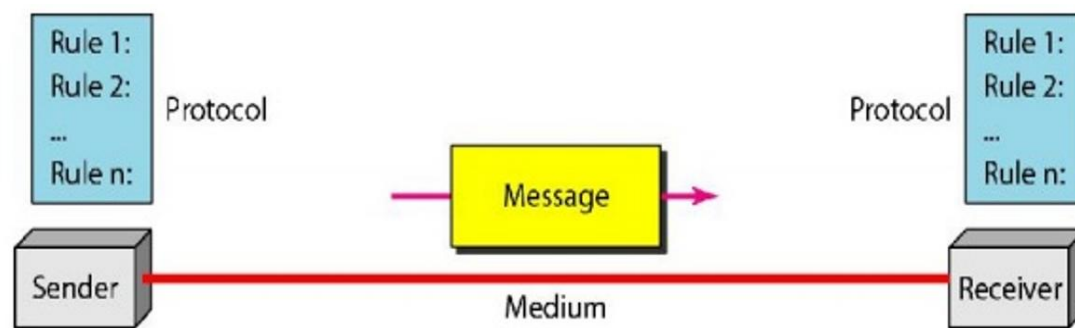


Sheet 1

1. Identify the five components of a data communications system. (use block diagram)

- **Message**
 - – The information to be transmitted (text, numbers, images, audio, video).
- **Sender**
 - – The device that generates and sends the message (e.g., computer, phone, camera).
- **Receiver**
 - – The device that receives the message (e.g., computer, TV, handset).
- **Transmission Medium**
 - – The physical path for communication (e.g., twisted-pair, coaxial, fiber optics, radio waves).
- **Protocol**
 - – A set of rules that ensure proper communication between devices.



2. What are the three criteria necessary for an effective and efficient network?

Criteria	Definition	Key Metrics / Factors
Performance	Ability of the network to deliver data efficiently.	- Transit time (delivery speed) - Response time (inquiry → reply) - Throughput (data transfer rate) - Delay (time taken) - Factors: users, medium, hardware, software
Reliability	Accuracy of data delivery over time.	- Frequency of failures - Recovery time after failure - Robustness during critical events
Security	Protection of data and systems against threats.	- Prevent unauthorized access - Prevent data damage/alteration - Data recovery policies & procedures

3. What are the advantages of a multipoint connection over a point-to-point connection?

Aspect	Types / Details	Advantages	Disadvantages
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Point-to-Point	Dedicated link between two devices (e.g., TV remote ↔ TV	full capacity for two devices	Point-to-point: costly with many devices
Multipoint (Multidrop)	Single link shared by multiple devices (spatially or time-shared)	efficient use of shared medium	performance drops if overloaded

4. Categorize the four basic topologies in terms of line configuration.

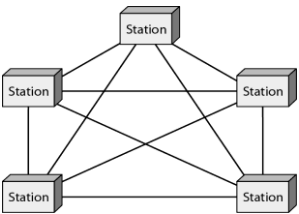
Topology	Line Configuration Type
Mesh Topology	Point-to-Point
Star Topology	Point-to-Point
Bus Topology	Multipoint
Ring Topology	Point-to-Point

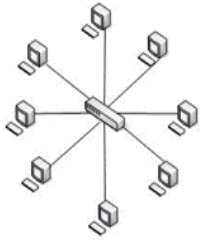
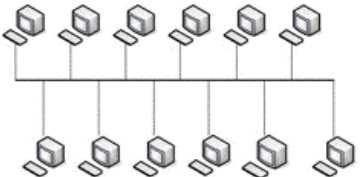
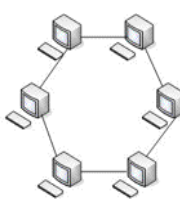
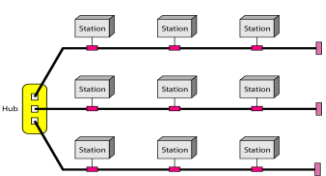
5. What is the difference between half-duplex and full-duplex transmission modes?

1. Half-Duplex
- Two-way communication, but not simultaneous (like a single-lane road).
 - At any moment, only one device can transmit while the other receives.
 - Examples: Walkie-talkies.
 - Channel capacity fully used in one direction at a time.
2. Full-Duplex
- Two-way communication simultaneously (like a two-way street).
 - Signals in both directions share the channel capacity.
 - Achieved by:
 - Two separate transmission paths, or
 - Splitting channel capacity.

Example: Telephone conversation

6. Name the four basic network topologies, and cite an advantage of each type.

Aspect	Definition	Advantages	Disadvantages	
Mesh Topology	Every device has a dedicated link to every other device	- High robustness - Privacy/security - Easy fault detection	- Very costly - Complex cabling - Requires many I/O ports	

Star Topology	All devices connected to a central hub	- Easy to install/reconfigure - Robust (one link failure doesn't affect others) - Easy fault isolation	- If hub fails (central device), whole system fails - More cabling than bus	
Bus Topology	One long backbone cable shared by devices	- Easy installation - Less cabling	- Hard to reconfigure - Fault in backbone stops all communication - Signal degradation	
Ring Topology	Devices connected in a closed loop, each device (node) linked to 2 neighbors	- Easy installation - Simplified fault isolation - Reconfigurable	- Break in ring disables network - Unidirectional traffic (unless dual ring used)	
Hybrid Topology	Combination of two or more topologies	- Flexible - Can leverage strengths of multiple topologies	- Complex design - Expensive	

7. For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?

Topology	Number of Cable Links	$n = 5$
Mesh Topology	$\frac{n(n-1)}{2}$	10
Ring Topology	n	5
Bus Topology	1	1
Star Topology	n	5

8. What is the Internet?

The Internet is a global network of interconnected computers and networks that communicate with each other using a common set of rules known as Internet Protocols (TCP/IP). It allows users all over the world to access, share, and exchange information, including text, images, videos, and other data.

9. Why are protocols needed?

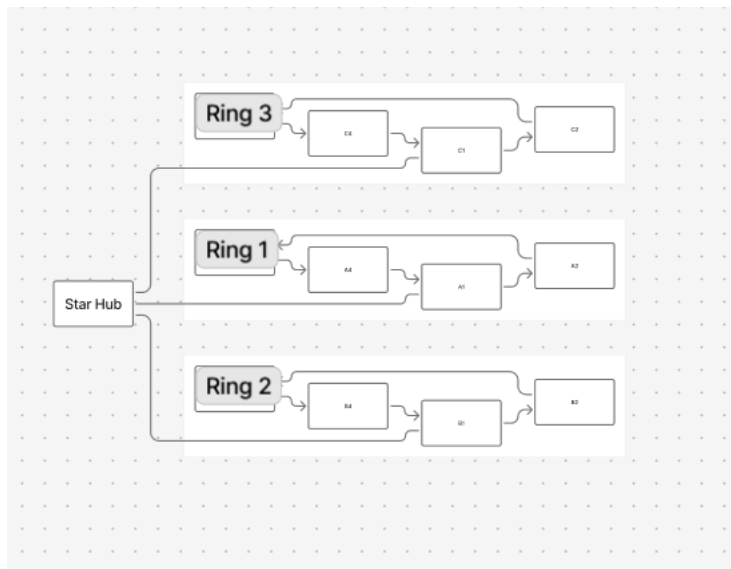
Protocols are needed because they define the rules and standards that allow computers and network devices to communicate effectively with each other.

Without protocols, devices from different manufacturers or systems would not understand one another, and data could not be exchanged correctly.

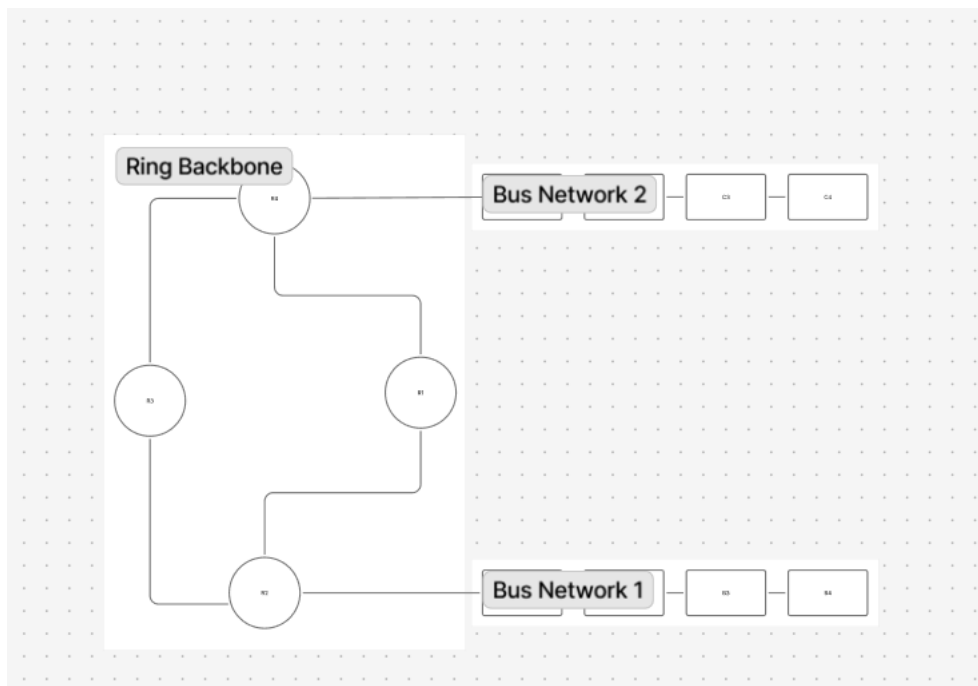
10. A color image uses 16 bits to represent a pixel. What is the maximum number of different colors that can be represented?

$$2^{16} = 65,536 \text{ colors}$$

11. Draw a hybrid topology with a star backbone and three ring networks.



12. Draw a hybrid topology with a ring backbone and two bus networks.



13. For each of the following four networks, discuss the consequences if a connection fails.

a) Five devices arranged in a mesh topology

In a mesh topology, every device is connected to every other device. Therefore, if one connection fails, the communication between the affected two devices will be interrupted, but all other devices can still communicate through alternate paths. The network remains fully operational, making it very reliable and fault tolerant.

b) Five devices arranged in a star topology (not counting the hub)

In a star topology, all devices are connected to a central hub or switch. If a connection between one device and the hub fails, only that particular device is affected, while the rest of the network continues to operate normally. However, if the central hub fails, the entire network will go down, as all communications pass through it.

c) Five devices arranged in a bus topology

In a bus topology, all devices share a single main communication cable called the backbone. If this main cable fails, the entire network will stop functioning because no device can communicate with another. However, if one device fails (without affecting the cable), the rest of the network remains unaffected. This makes the bus topology less reliable.

d) Five devices arranged in a ring topology

In a ring topology, each device is connected to exactly two other devices, forming a closed loop. If one connection fails, the entire network is typically disrupted because the loop is broken and data cannot

circulate. However, in a dual-ring system, communication can continue in the opposite direction, providing better reliability.

Multiple Choice Questions

1. Communication between a computer and a keyboard involves _____ transmission
 - (a) Full-duplex
 - (b) Half-duplex
 - (c) Simplex**
 - (d) None of these
2. A _____ is the physical path over which a message travels .
 - a) Path
 - b) Medium**
 - c) Protocol
 - d) Route
3. A _____ set of rules that governs data communication.
 - a) Protocols**
 - b) Standards
 - c) RFCs
 - d) Servers
4. Three or more devices share a link in _____ connection.
 - a) Unipoint
 - b) Multipoint**
 - c) Point to point
 - d) Simplex
5. In computer network nodes are _____
 - a) the computer that originates the data
 - b) the computer that routes the data
 - c) the computer that terminates the data
 - d) all of the mentioned**
6. In which transmission mode is data sent in only one direction?
 - a) Half-duplex
 - b) Full-duplex
 - c) Simplex**
 - d) Multiplexing
7. Which of the following is an example of a device that operates in simplex mode?
 - a) Walkie-talkie
 - b) Keyboard to monitor communication**
 - c) Telephone
 - d) Ethernet network
8. Half-duplex mode allows for communication in:

a) Both directions, but not simultaneously

b) Both directions, simultaneously

c) Only one direction

d) Multiple directions at once

9. Which of the following applications uses half-duplex transmission mode?

a) Telephone conversation

b) Radio broadcasting

c) Walkie-talkie communication

d) Fiber optic network

10. Full-duplex mode enables:

a) Data transmission in only one direction

b) Transmission and reception of data simultaneously

c) Alternating data transmission in both directions

d) Data transmission in short bursts

11. Which of the following applications operates in full duplex mode?

a) Radio broadcasting

b) Walkie-talkies

c) Telephone systems

d) Television broadcasting

12. Which transmission mode is ideal for real-time applications such as video calls?

a) Simplex

b) Half-duplex

c) Full-duplex

d) None of the above

13. Which type of network communication commonly uses full-duplex mode?

a) Broadcast radio

b) Ethernet networks

c) Morse code transmission

d) Satellite communication